

HW 13/07/26

- 1) Write a Cpp code which includes a class student.
Find out the space occupied by the object
by using `sizeof()`.

```

1) #include <iostream>
using namespace std;
class Student
{
public:
    int roll; // 4 bytes
    char grade; // 1 byte
    float marks; // 4 bytes
};
int main()
{
    Student s1;
    cout << "Size of student class object is " <<
        sizeof(s1) << " bytes";
}

```

Output : Size of student class object is 12 bytes

Actually total = 9 bytes only but compiler adds padding, so `sizeof()` function gives us 12 bytes. A float usually needs to start at an address that is a multiple of 4. (i.e.) $[4 + 1 + 3 + 4 = 12]$.

If we add member functions then size of the object does not increase because member functions are shared by all the objects of the class.

Even if we define the functions outside the class, the object size remains the same.